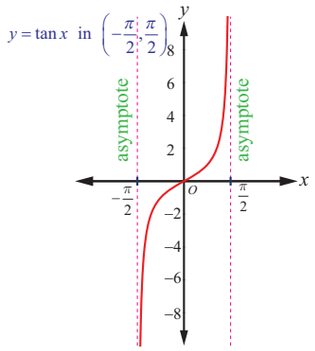


Fig. 4.17

x	y
-1	$-\frac{\pi}{4}$
0	0
$-\frac{\sqrt{3}}{3}$	$-\frac{\pi}{6}$
1	$\frac{\pi}{4}$

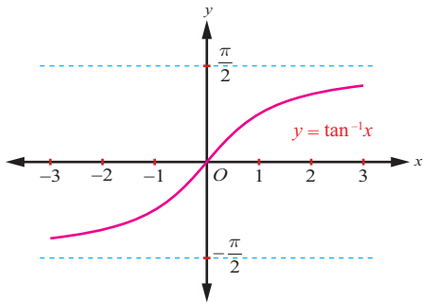


Fig. 4.18

Note

(i) The inverse tangent function is strictly increasing and continuous on the domain $(-\infty, \infty)$.

(ii) The graph of $y = \tan^{-1} x$ passes through the origin.

(iii) The graph is symmetric with respect to origin and hence, $y = \tan^{-1} x$ is an odd function.

Example 4.8

Find the principal value of $\tan^{-1}\left(\sqrt{3}\right)$.

Solution

Let $\tan^{-1}\left(\sqrt{3}\right) = y$. Then, $\tan y = \sqrt{3}$. Thus, $y = \frac{\pi}{3}$. Since $\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Thus, the principal value of $\tan^{-1}\left(\sqrt{3}\right)$ is $\frac{\pi}{3}$.

Example 4.9

Find (i) $\tan^{-1}\left(-\sqrt{3}\right)$ (ii) $\tan^{-1}\left(\tan\frac{3\pi}{5}\right)$ (iii) $\tan\left(\tan^{-1}(2019)\right)$

Solution

(i) $\tan^{-1}\left(-\sqrt{3}\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{3}\right)\right) = -\frac{\pi}{3}$, since $-\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(ii) $\tan^{-1}\left(\tan\frac{3\pi}{5}\right)$.

Let us find $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\tan\theta = \tan\frac{3\pi}{5}$.

Since the tangent function has period π, $\tan\frac{3\pi}{5} = \tan\left(\frac{3\pi}{5} - \pi\right) = \tan\left(-\frac{2\pi}{5}\right)$.

Therefore, $\tan^{-1}\left(\tan\frac{3\pi}{5}\right) = \tan^{-1}\left(\tan\left(-\frac{2\pi}{5}\right)\right) = -\frac{2\pi}{5}$, since $-\frac{2\pi}{5} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(iii) Since $\tan\left(\tan^{-1} x\right) = x, x \in \mathbb{R}$, we have $\tan\left(\tan^{-1}(2019)\right) = 2019$.

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